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Third Semester –B.Tech-Course work

MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code- ICMTC04/EEMTC04

Course Title- Mathematics III

Time :1:30 Hours

Max. Marks :25

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer

Q.No	Question	Marks	CO
1a	Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^2+y^2}$ are harmonic functions of (x,y) but are not harmonic conjugates.	2.5	CO1
1b	Find the imaginary part of the analytic function whose real part is $x^3 - 3xy^2 + 3x^2 - 3y^2$	2.5	CO1
2a	State and prove Cauchy's Integral Theorem.	2.5	CO1
2b	Evaluate the following integral using Cauchy's Integral Formula $\int_c \frac{\log z}{(z-1)^3} dz$ where c is $ z-1 = \frac{1}{2}$	2.5	CO1
3a	Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos \theta} d\theta$ by using Contour Integration .	2.5	CO1
3b	If $\vec{a}$ is a constant vector and $\vec{r}$ the radius vector prove that $\nabla \left(\frac{\vec{a} \cdot \vec{r}}{r^n} \right) = \frac{\vec{a}}{r^n} - \frac{n(\vec{a} \cdot \vec{r})\vec{r}}{r^{n+2}}$	2.5	CO2
4a	$\nabla^2 r^n = n(n+1)r^{n-2}$ where $r = \vec{r} $	2.5	CO2
4b	Find the values of constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1, 2, -1)$ has maximum of magnitude 64 in a direction parallel to the z axis.	2.5	CO2
5a	A vector field is given by $\vec{F} = (y \sin z - \sin x)\vec{i} + (x \sin z + 2yz)\vec{j} + (xy \cos z + y^2)\vec{k}.$ Show that the field is irrotational and find the scalar potential.	2.5	CO2
5b	A vector field $\vec{F} = x^2\vec{i} - xy\vec{j}$. Find the line integral from origin O to the point $P(1,1)$ along the x axis from O to $A(1,0)$ and then from $A(1,0)$ to $P(1,1)$	2.5	CO2